

PYTHON FOR DATA ANALYTICS 201

Capstone project

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09/30/26

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WATER, WATER, EVERYWHERE...

AN EXPLORATORY DATA ANALYSIS OF WATER QUALITY AT NYC BEACHES

DATA SET: NYC OPEN DATA BEACH WATER SAMPLES

MOTIVATION:

- Interest in nature and environmental data
- Interest in environmental impact such as heavy rainfall and flooding
- Water quality and safety is a public health issue
- NYC beaches saw more than 15 million visitors in 2026

ABOUT THE DATA SET:

- Samples collected by Beach Surveillance Program April - Sept.
- Sampling frequency is weekly at all beaches except Rockaway and Breezy Point (bi-weekly)
- Data set encompasses 31.2k rows of data
- Samples measure Enterococci bacteria in beach water samples
- Measurements are in Most Probable Number / 100mL (MPN/100mL)
- 100 mL = approximately 3.38 fluid oz.
- A sample resulting in greater than 104MPN/100mL is the "exceedance threshold"

WHAT THE LEVELS MEAN:

WARNING



Swimming and wading are **not recommended.**

Water is contaminated with sewage or storm runoff, which may cause vomiting, diarrhea, respiratory illness or infections. Children, pregnant women, the elderly and the chronically ill are at higher risk.

For beach status updates:

Text **BEACH** to 877-877 or call 311



BEACH ADVISORY

- Single water sample exceeds 104MPN/100mL
- Swimming and wading not recommended

BEACH CLOSURE

- An amount exponentially higher than 104 MPN/mL is detected
- Consecutive tests results of 104 MPN/mL
- 30-day geometric mean > 35 MPN/100mL
- Swimming and wading are not permitted

BEACH CLOSED



By order of the Health Department,

Swimming and wading are **not permitted.**

For beach status updates:

Text **BEACH** to 877-877 or call 311



RESEARCH QUESTIONS:

- What are the descriptive statistics for the total amount of Enterococci bacteria
 - What about for observations > 104 MPN/100mL (exceedance threshold)
- What Borough and beach recorded the most exceedance
- What year showed greatest exceedance
- Has overall exceedance level increased or decreased over time
- What did the summer of 2026 look like in relation to the overall data set
- Is there a 'cleanest' beach

DATA CLEANING/TRANSFORMATIONS:

- **Converted and extracted** 'Sample Date' to useable datetime data type
- **Dropped** 4 beaches with a lot of incomplete data (many missing years)
- **Merged** several 'Beach Name' labels to eliminate duplicates: 48 —> 36 beaches
- **Tested nulls & imputed values:**
 - 'Enterococci Bacteria' had 7464 NaNs, each represented an 'undetectable amount' of less than 1 MPN/100mL. After research and testing, I decided to use the EPA guideline of half the undetectable value, 0.5 in this case
 - 'Sample Location' refers to Left, Right or Center of beach; had 37 NaNs, which I imputed with 'Center,' and will not skew this particular analysis
- **Limited date range** to 2011-2026 for most complete data set (several beaches had missing data in earlier years)
- **Added geography** by creating a second CSV containing Latitude, Longitude, Borough, and whether beach was Public_Private; joined it with the NYC data set
- **Aggregated** cleaned data set to average multiple samples per beach per day into one observation per beach per day
- **Libraries** pandas, geopandas, matplotlib, seaborn, numpy

OVERVIEW OF DATA SET:

CLEANED DATA FRAME

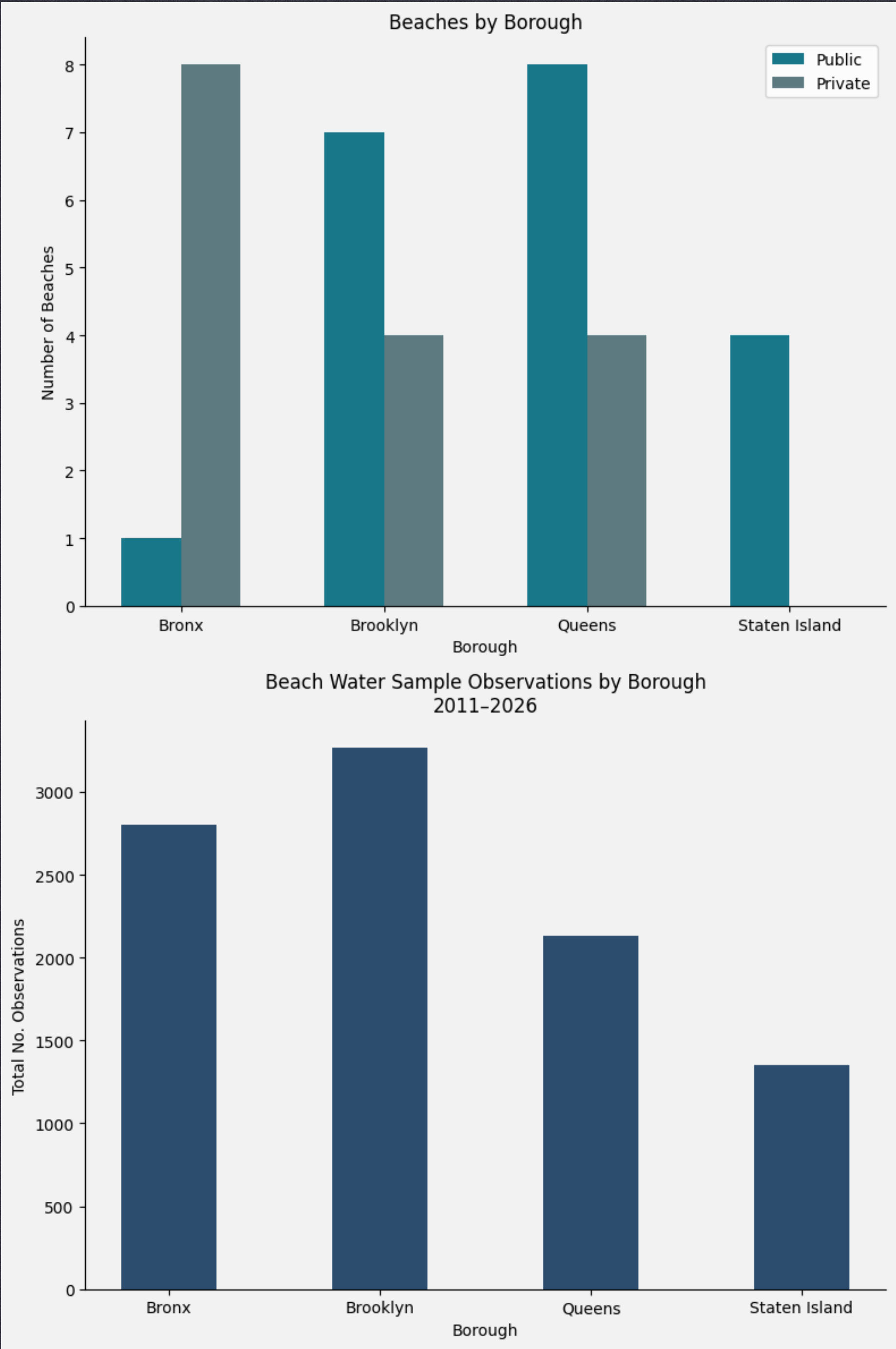
...

	Sample ID	Sample Date	Beach Name	Sample Location	Enterococci Results	Units or Notes	Year	Latitude	Longitude	Borough	Public_Private
0	JB2609141040-1.1	2026-09-14	SCHUYLER HILL CIVIC ASSOCIATION	Left	10.0	MPN/100 ml	2026	40.8174	-73.8123	Bronx	private
1	KB2609140940-1.1	2026-09-14	SEAGATE 38TH	Left	9.9	MPN/100 ml	2026	40.5750	-74.0080	Brooklyn	private
2	JB2609141005-1.2	2026-09-14	WEST FORDHAM STREET ASSOCIATION	Center	31.0	MPN/100 ml	2026	40.8475	-73.7845	Bronx	private
3	KB2609140940-1.3	2026-09-14	SEAGATE 38TH	Right	20.0	MPN/100 ml	2026	40.5750	-74.0080	Brooklyn	private
4	JB2609141005-1.3	2026-09-14	WEST FORDHAM STREET ASSOCIATION	Right	10.0	MPN/100 ml	2026	40.8475	-73.7845	Bronx	private



OVERVIEW OF DATA SET:

BEACH AND OBSERVATION TOTALS BY BOROUGH



DESCRIPTIVE STATISTICS:

#what are the descriptive stats

df3['Ent SampleAvg'].describe()

...	Ent SampleAvg
count	9551.000000
mean	65.277837
std	416.059325
min	0.500000
25%	3.000000
50%	9.900000
75%	22.666667
max	24196.000000

dtype: float64

Approximately 75% of observations had Enterococci concentrations at or below 22.67 MPN/100ML

17% of total observations tested undetectable at < 1 MPN/100mL

	0
2011	173
2012	123
2013	154
2014	205
2015	235
2016	190
2017	135
2018	172
2019	160
2020	102
2026	5
Total	1654

dtype: int64

...

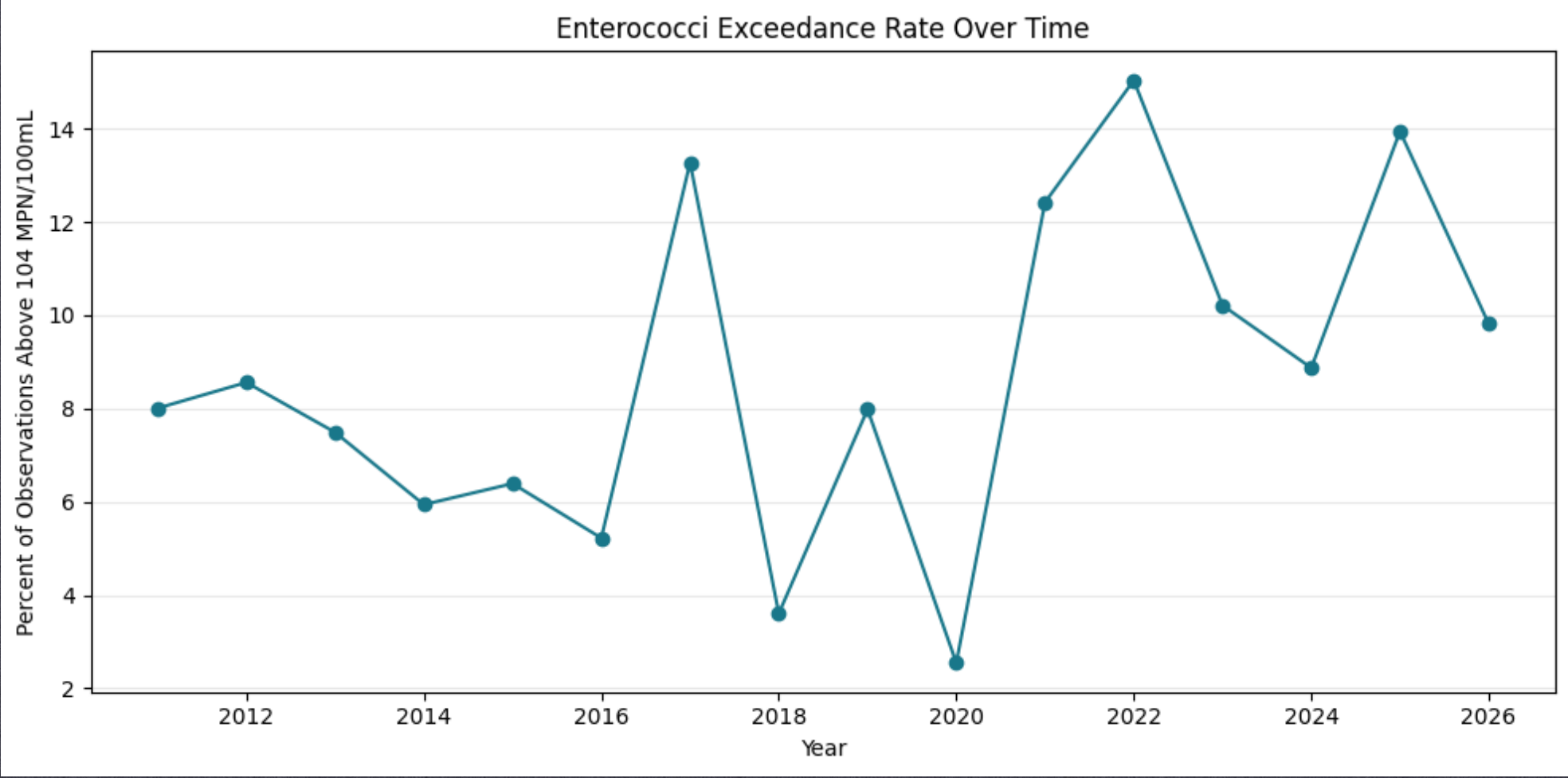
2276

Beach Name	TRINITY DANISH YOUNG PEOPLES SOCIETY
Borough	Bronx
Public_Private	private
Sample Date	2025-06-17 00:00:00
Sample Location	Right
Enterococci Results	24196.0

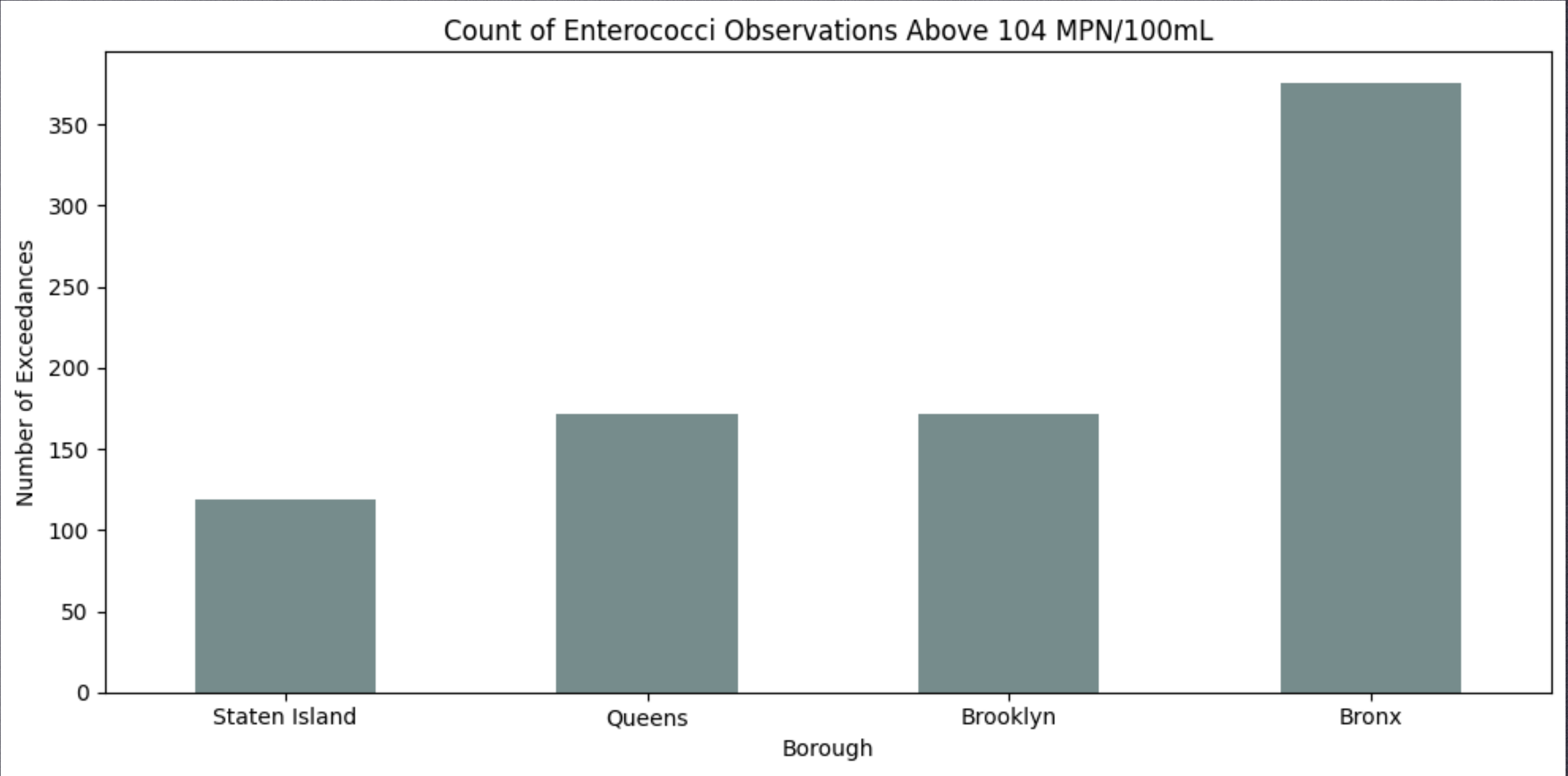
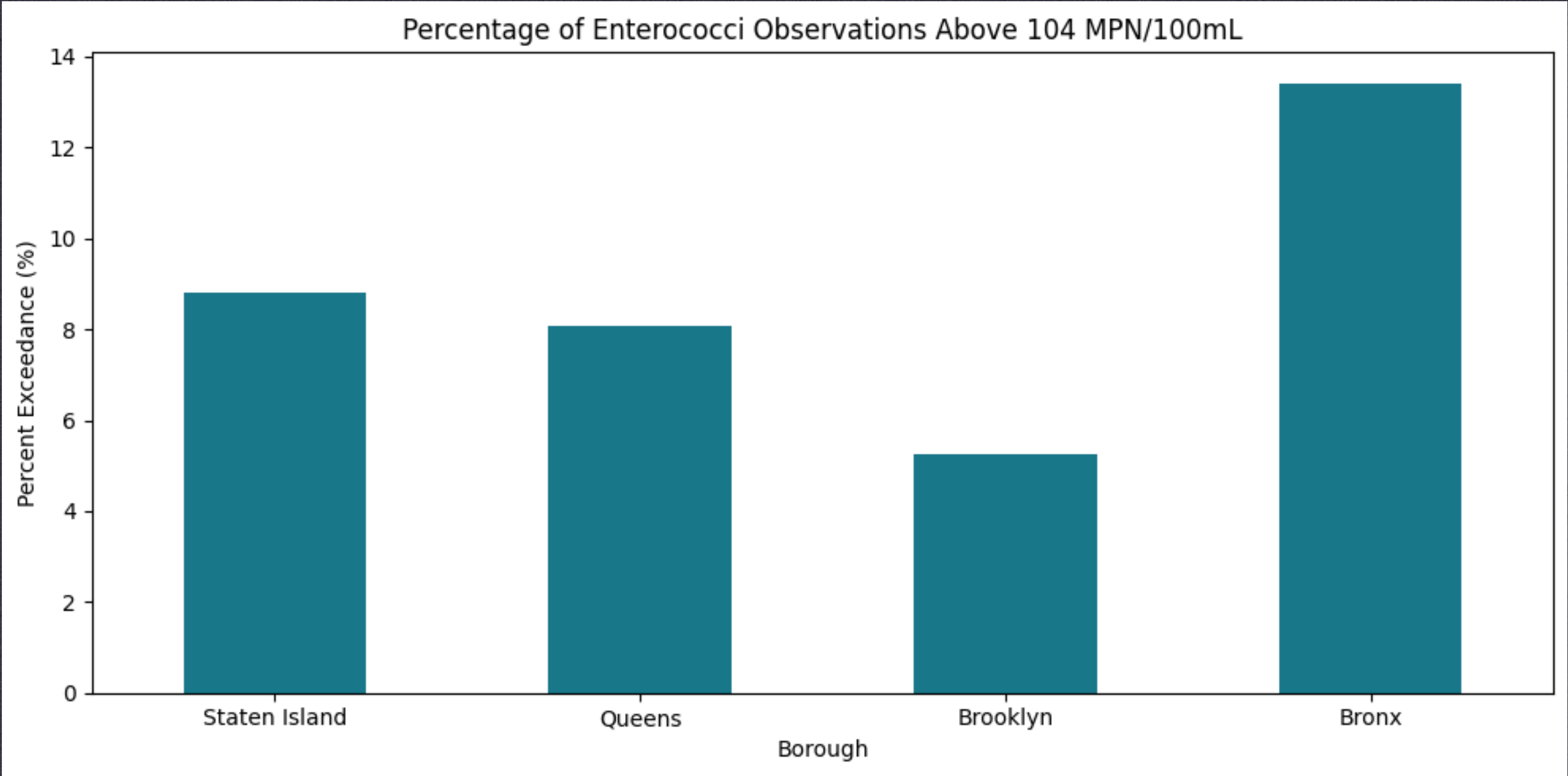
dtype: object

The maximum concentration of 24,196 MPN/100mL occurred on 6/17/2025, at the private beach, Trinity Danish Young People’s Society in the Bronx

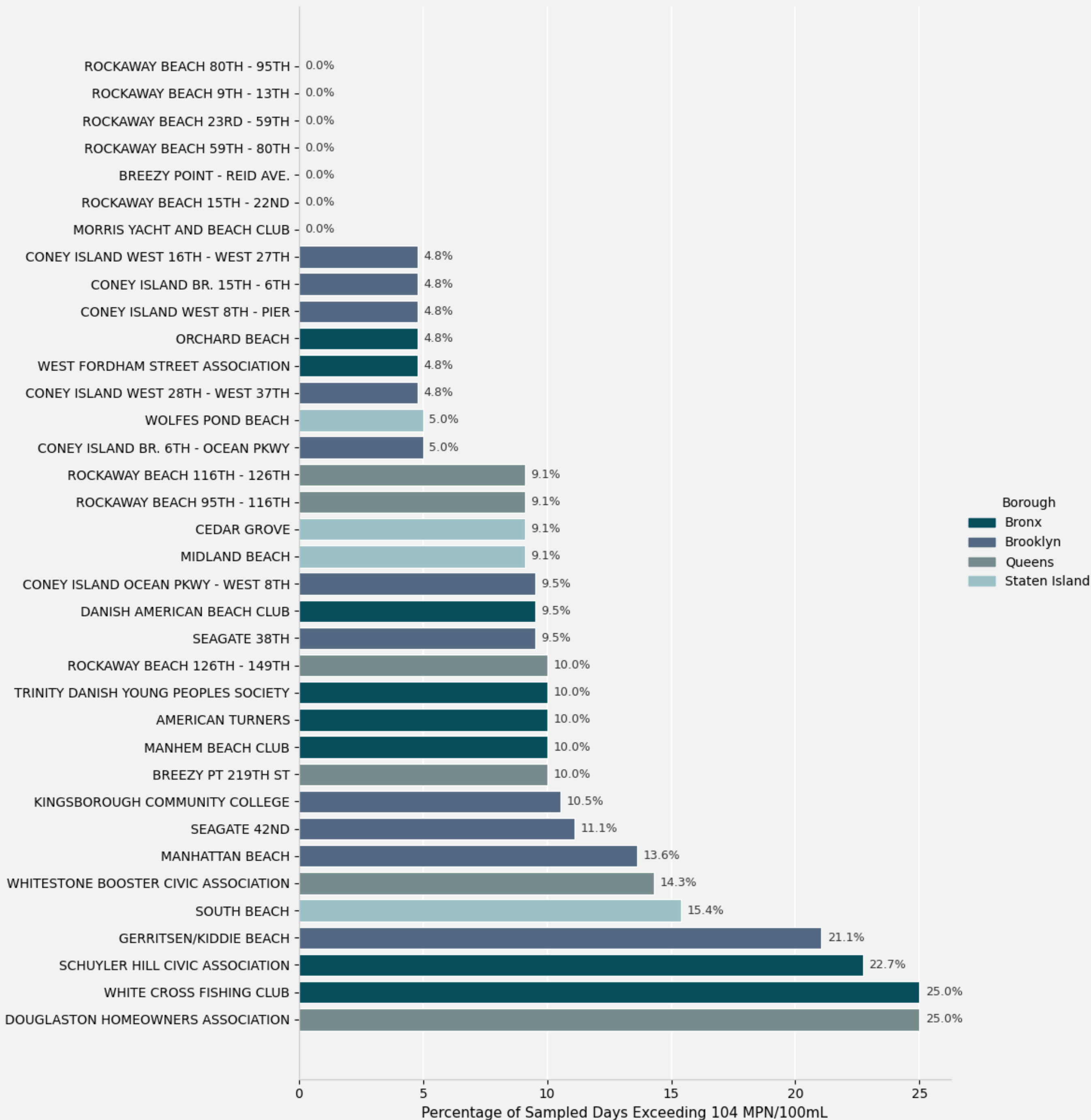
EXCEEDANCE EVENTS:



- **2022** observations showed greatest exceedance
- **2018** observations showed least exceedance (excluding 2020)
- **BRONX** experienced greatest percentage and count of exceedance
- **TREND** - exceedance seems to be increasing over time?



Percentage of Days Above 104 MPN/100mL by Beach, 2026



2026 SNAPSHOT:

- **5 UNDETECTABLE** observations < 0.5 MPN/100mL
- **637 DETECTABLE** observations > 0.5 MPN/100mL
- **61 EXCEEDANCE** observations > 104 MPN/100mL
- **9.5% EXCEEDANCE** rate overall

Rockaways and Breezy Point beaches rate among lowest exceedance

Douglaston Homeowners Assc. And White Cross Fishing Club beaches rate among highest exceedance

Percentage of Days Above 104 MPN/100mL by Beach and Year

Beach																	
	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	
Bronx	AMERICAN TURNERS	18.2	21.7	20.0	15.0	14.3	10.0	55.0	0.0	19.0	0.0	31.2	33.3	21.1	11.1	33.3	10.0
	DANISH AMERICAN BEACH CLUB	4.8	17.4	5.0	19.0	9.5	9.5	45.0	0.0	28.6	0.0	12.5	31.6	19.0	15.8	36.8	9.5
	MANHEM BEACH CLUB	9.5	13.0	5.0	10.0	4.8	10.0	30.0	0.0	19.0	0.0	13.3	16.7	10.5	5.6	27.8	10.0
	MORRIS YACHT AND BEACH CLUB	20.0	12.0	4.3	0.0	0.0	0.0	5.3	0.0	4.3	0.0	5.6	15.8	4.8	0.0	10.0	0.0
	ORCHARD BEACH	0.0	10.0	0.0	0.0	4.5	4.8	4.5	0.0	0.0	7.7	9.5	10.0	4.8	5.0	0.0	4.8
	SCHUYLER HILL CIVIC ASSOCIATION	9.1	10.5	0.0	5.3	0.0	5.3	20.0	0.0	15.0	0.0	6.2	9.5	14.3	21.1	18.2	22.7
	TRINITY DANISH YOUNG PEOPLES SOCIETY	22.7	23.8	10.5	10.5	9.5	10.5	36.8	0.0	19.0	0.0	25.0	33.3	19.0	15.8	38.9	10.0
	WEST FORDHAM STREET ASSOCIATION	4.5	19.0	13.6	4.3	8.7	4.5	20.0	0.0	0.0	0.0	0.0	26.3	4.5	0.0	5.0	4.8
	WHITE CROSS FISHING CLUB	14.3	17.4	14.3	15.0	21.7	15.8	47.4	5.0	23.8	0.0	46.7	38.9	26.3	27.8	42.1	25.0
Brooklyn	CONEY ISLAND BR. 15TH - 6TH	5.0	0.0	0.0	5.0	9.5	4.8	4.8	0.0	5.0	0.0	0.0	0.0	0.0	0.0	5.3	4.8
	CONEY ISLAND BR. 6TH - OCEAN PKWY	0.0	0.0	4.8	0.0	9.1	4.8	0.0	0.0	0.0		11.1	0.0	0.0	0.0	5.3	5.0
	CONEY ISLAND OCEAN PKWY - WEST 8TH	0.0	0.0	5.0	5.0	4.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.5
	CONEY ISLAND WEST 16TH - WEST 27TH	0.0	5.3	4.8	0.0	0.0	9.5	4.8	0.0	0.0		11.1	15.0	4.5	5.3	0.0	4.8
	CONEY ISLAND WEST 28TH - WEST 37TH	0.0	5.3	4.8	5.0	4.8	4.8	4.8	5.0	5.0	0.0	9.1	0.0	0.0	0.0	5.3	4.8
	CONEY ISLAND WEST 8TH - PIER	4.0	5.3	5.0	0.0	4.8	0.0	0.0	0.0	0.0		11.1	15.0	18.2	0.0	10.5	4.8
	GERRITSEN/KIDDIE BEACH	13.6	10.5	15.0	5.0	0.0	15.0	15.0	15.0	20.0	12.5	6.7	16.7	23.8	27.8	5.6	21.1
	KINGSBOROUGH COMMUNITY COLLEGE	4.8	0.0	18.2	9.5	9.5	10.5	15.0	0.0	5.0	0.0	26.7	0.0	0.0		30.0	10.5
	MANHATTAN BEACH	4.8	0.0	0.0	5.0	0.0	4.8	8.7	0.0	5.0	7.1	10.5	20.0	5.0	5.0	9.5	13.6
Queens	SEAGATE 38TH	9.5	5.3	0.0	5.0	4.8	0.0	5.0	0.0	0.0	0.0	6.7	0.0	0.0	5.6	5.6	9.5
	SEAGATE 42ND	7.1	5.3	0.0	5.0	4.5	0.0	5.0	0.0	0.0	0.0	0.0	0.0	5.3	0.0	5.6	11.1
	BREEZY POINT - REID AVE.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	BREEZY PT 219TH ST	21.4	7.7	0.0	0.0	9.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11.1	10.0
	DOUGLASTON HOMEOWNERS ASSOCIATION	34.8	17.4	38.5	25.0	21.7	13.6	35.0	13.6	27.3	22.2	50.0	47.4	31.8	36.8	30.0	25.0
	ROCKAWAY BEACH 116TH - 126TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.1
	ROCKAWAY BEACH 126TH - 149TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0
	ROCKAWAY BEACH 15TH - 22ND	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.7	0.0	0.0	0.0	11.1	0.0
	ROCKAWAY BEACH 23RD - 59TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Staten Island	ROCKAWAY BEACH 59TH - 80TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	ROCKAWAY BEACH 80TH - 95TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	ROCKAWAY BEACH 95TH - 116TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.1
	ROCKAWAY BEACH 9TH - 13TH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	WHITESTONE BOOSTER CIVIC ASSOCIATION	9.1	4.8	4.8	0.0	10.0	5.0	20.0	5.0	25.0	12.5	13.3	38.9	40.0	36.8	36.8	14.3
	CEDAR GROVE	3.3	5.3	16.7	9.5	12.5	4.8	9.1	20.8	5.0	0.0	9.1	5.0	13.6	10.0	10.0	9.1
	MIDLAND BEACH	6.7	5.3	0.0	4.8	0.0	0.0	4.8	4.8	5.0	0.0	5.0	5.0	4.8	0.0	10.0	9.1
	SOUTH BEACH	15.6	5.3	5.0	4.8	8.7	4.5	4.8	12.5	0.0	7.1	8.7	0.0	21.7	14.3	20.0	15.4
	WOLFES POND BEACH	8.7	10.0	19.0	4.8	9.1	5.0	4.8	0.0	5.0	0.0	20.8	28.0	0.0	15.0	19.0	5.0

CONCLUSIONS:

- Exceedance at NYC beaches appears to be increasing over time
- 100% detectible bacteria rate in samples 2021-2025
- 2026 showed improvement from 2025, but still slightly higher overall:
 - exceedance (9.5%) was 1.3% above the mean, although it was an improvement from the 2025 rate (14%)
 - had 5 undetectable observations, also slight improvement from years prior
- Highest exceedance occurs most consistently at Douglaston Homeowner's Assc. and White Cross Fishing Club beaches
- Lowest exceedance occurs most consistently at Rockaways and Breezy Point beaches
- Geography appears to be more of a determining factor than whether a beach is private or public (proximity to waste water, sewer outflows)

RECOMMENDATIONS / FURTHER ANALYSIS:

- Increase sampling frequency at all beaches
- Sample Rockaways and Breezy Point with the same frequency as the other beaches
- Investigate further the influence of rainfall, flooding, and combined sewer overflows on bacteria levels
- Investigate relationship between wastewater treatment plants and bacterial pollution

REFERENCES:

DOHMC data collection process:

https://data.cityofnewyork.us/Health/Beach-Water-Samples/2xir-kwzz/about_data

DOHMC water quality map with most recent samples:

<https://a816-dohbsp.nyc.gov/IndicatorPublic/Beaches/>

NYCEP combined sewer overflows:

<https://www.nyc.gov/site/dep/water/combined-sewer-overflows.page>

NYC Open Data Beach Water Samples:

https://data.cityofnewyork.us/Health/Beach-Water-Samples/2xir-kwzz/about_data

NYC Open Data Floodnet Data:

https://data.cityofnewyork.us/Environment/FloodNet-Street-Flooding-Events-Measured-by-FloodN/aq7i-eu5q/about_data

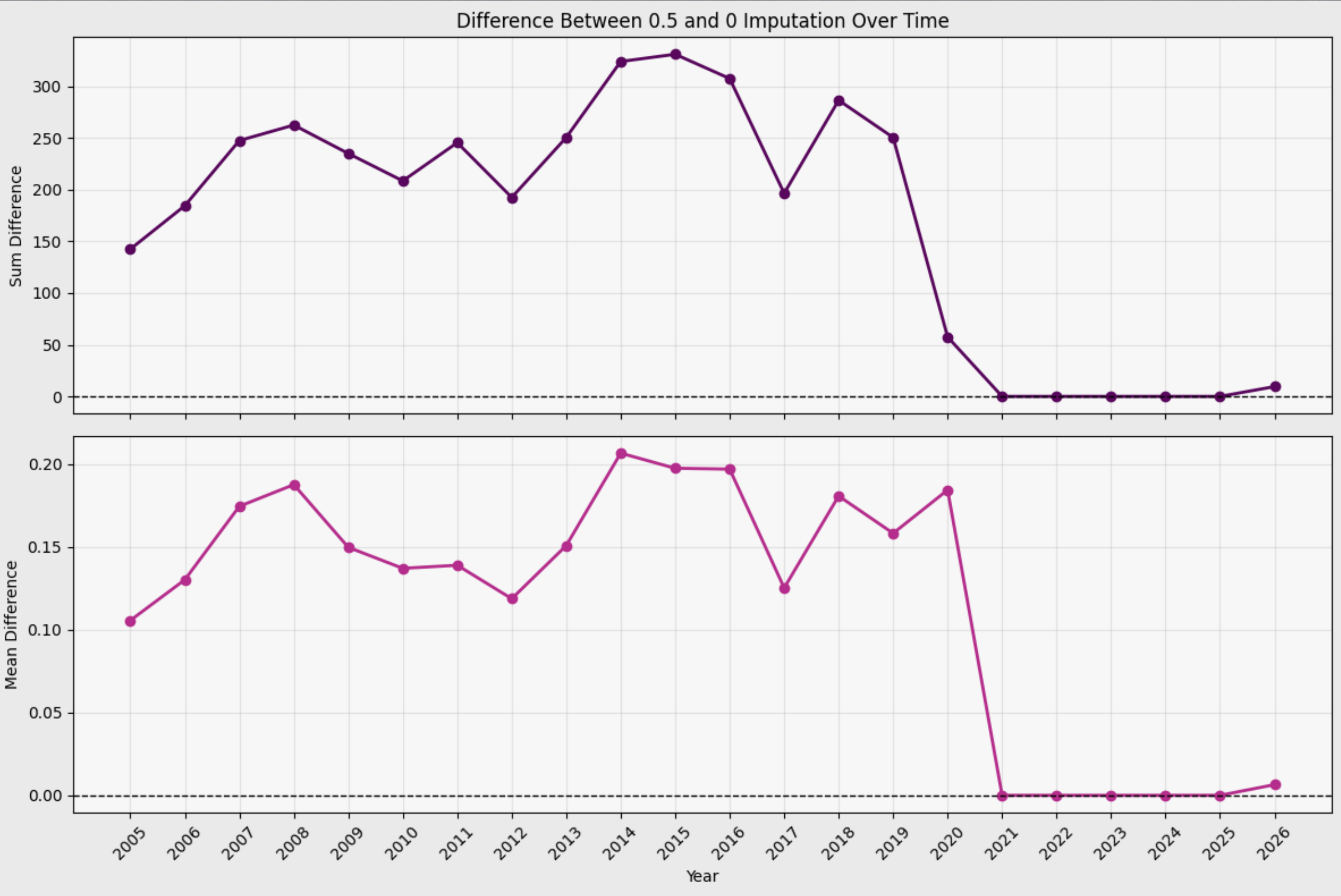
Office of the Mayor

<https://www.nyc.gov/mayors-office/news/2026/09/mayor-mamdani-announces-more-than-16-million-people-visited-new->

ADDITIONAL INFO: COMPARISON OF IMPUTATION METHODS:

Difference between 0.5 and 0 imputation

	Statistic	point5 Imputation	Zero Imputation	Difference (point5 - Zero)
0	Sum	2.184799e+06	2.181067e+06	3732.000000
1	Mean	7.009301e+01	6.997328e+01	0.119731
2	Median	9.900000e+00	9.900000e+00	0.000000



Sum difference and mean difference of 0.5 and 0 imputation expressed over time

ADDITIONAL INFO: HIGH CONCENTRATION EVENTS

...

	Ent SampleAvg
count	31.000000
mean	5246.494624
std	4170.908571
min	2533.333333
25%	3389.166667
50%	3931.000000
75%	4836.000000
max	24196.000000
dtype: float64	

9	2025-07-15	DANISH AMERICAN BEACH CLUB	4570.0	Bronx
19	2025-07-15	DOUGLASTON HOMEOWNERS ASSOCIATION	3542.3333333333335	Queens
22	2025-07-15	MANHEM BEACH CLUB	3389.3333333333335	Bronx
0	2025-06-17	TRINITY DANISH YOUNG PEOPLES SOCIETY	24196.0	Bronx
21	2025-06-16	MANHATTAN BEACH	3420.0	Brooklyn
2	2025-05-12	MIDLAND BEACH	8214.0	Staten Island
3	2024-07-23	AMERICAN TURNERS	7731.666666666667	Bronx
4	2024-07-23	TRINITY DANISH YOUNG PEOPLES SOCIETY	7447.666666666667	Bronx
6	2024-07-23	WHITE CROSS FISHING CLUB	5800.0	Bronx
11	2023-08-15	AMERICAN TURNERS	4335.333333333333	Bronx
14	2023-08-15	TRINITY DANISH YOUNG PEOPLES SOCIETY	3963.6666666666665	Bronx
15	2023-08-15	DOUGLASTON HOMEOWNERS ASSOCIATION	3931.0	Queens
18	2023-08-15	DANISH AMERICAN BEACH CLUB	3678.0	Bronx
23	2023-08-15	WHITE CROSS FISHING CLUB	3389.0	Bronx
24	2023-08-15	MANHEM BEACH CLUB	3323.0	Bronx
10	2023-08-07	CONEY ISLAND WEST 8TH - PIER	4352.0	Brooklyn
25	2023-08-07	GERRITSEN/KIDDIE BEACH	3152.6666666666665	Brooklyn
29	2023-08-07	SOUTH BEACH	2736.6666666666665	Staten Island

For bacteria concentrations over 2500MPN/100mL, there are 4 dates with apparent spikes

- 2023-08-07 - 3 beaches in BK and SI
- 2023-08-15 - 6 beaches in QNS and BX
- 2024-07-23 - 3 beaches in BX
- 2025-07-15 - 3 beaches in QNS and BX